About This Dashboard

This dashboard analyzes the short-term performance, risk characteristics, and diversification potential of selected technology stocks. The analysis is based on daily market data and focuses on understanding how these assets behave individually and relative to each other within a portfolio.

All calculations are performed using a **30-day rolling window**, ensuring that performance, volatility, and correlation metrics are directly comparable across all visualizations. This time horizon provides a consistent view of recent market dynamics while capturing short-term trends and risk behavior.

Several key metrics are used throughout the dashboard. **30-Day Return** measures the percentage change in price over the last 30 trading days and represents short-term performance. **30-Day Volatility** is calculated as the standard deviation of daily returns over the same period and serves as a proxy for risk. **Risk-Adjusted Return** divides return by volatility to show how efficiently each stock generates performance relative to the level of risk taken.

The **Correlation Matrix** highlights how strongly the selected stocks move together. Lower correlations indicate stronger diversification potential, as assets that behave differently can reduce overall portfolio risk when combined.

This project demonstrates a typical analytical workflow used in financial data analysis, including data transformation, rolling calculations, and portfolio analytics. The data processing and calculations are performed in **Google BigQuery using SQL window functions**, while the visualizations are built in **Looker Studio** to provide an interactive and intuitive analytical dashboard.

Data Pipeline Overview:

1. **Data Ingestion** – Daily stock price data is retrieved via a financial market API (Alpha Vantage) and loaded into Google BigQuery. A Cloud Run service retrieves and updates the data daily.
2. **Data Transformation** – SQL queries and window functions are used to calculate rolling returns, volatility, moving averages, and portfolio metrics.
3. **Feature Engineering** – Additional indicators such as risk-adjusted return and correlation values are derived for deeper analysis.
4. **Visualization Layer** – The processed datasets are connected to Looker Studio to create interactive dashboards for performance, risk, and diversification analysis.

Created by Tófalvi Richárd, Data Analytics Portfolio Project

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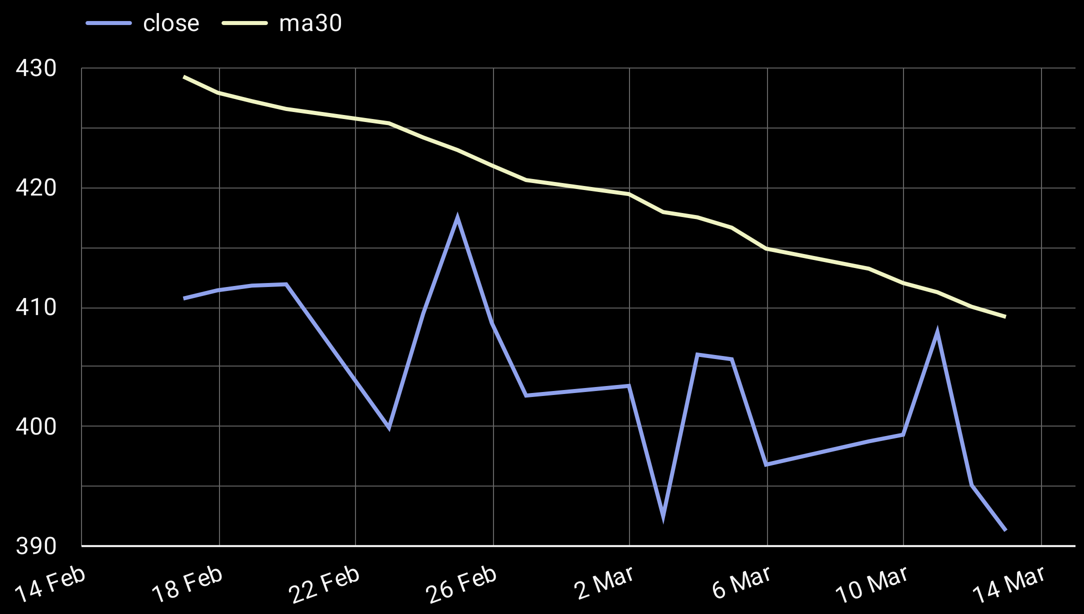
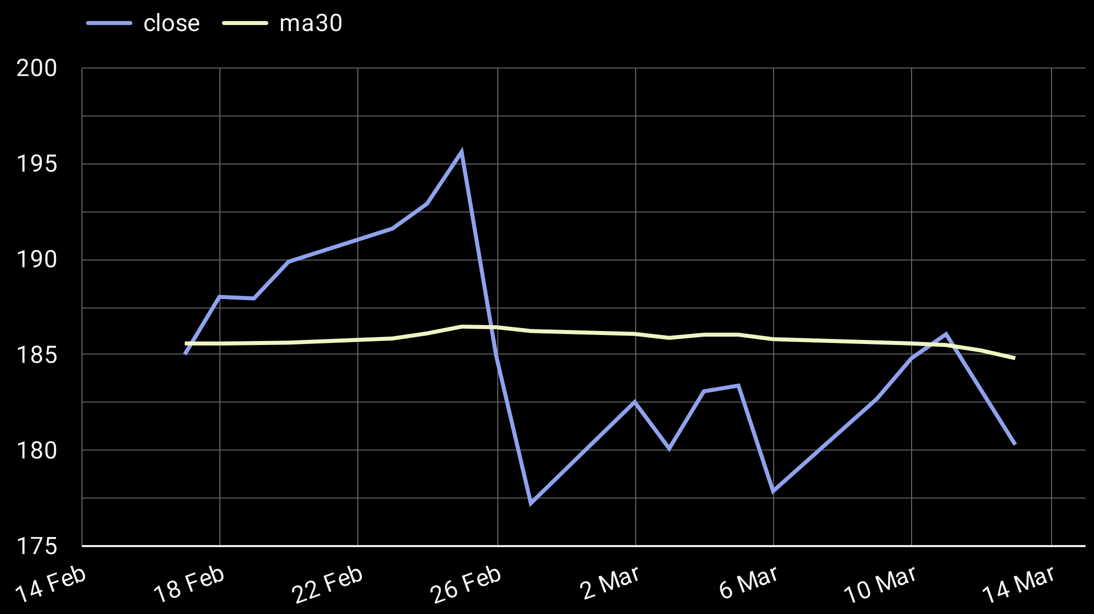
30-Day Trend Analysis – Big Tech

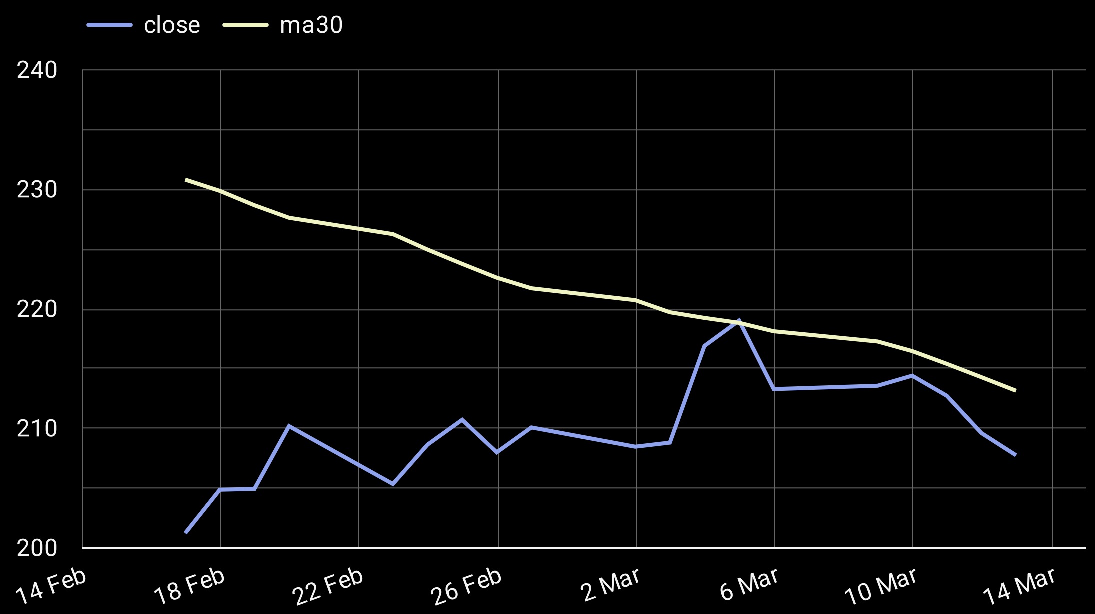
Tesla(TSLA)



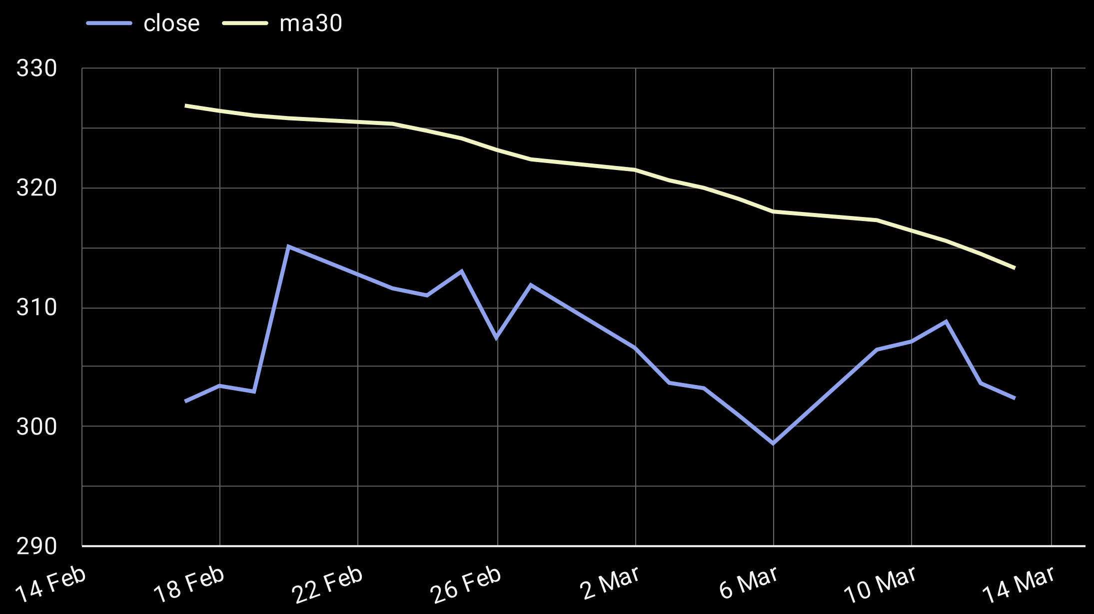
Nvidia(NVDA)



Amazon(AMZN)



Google(GOOGL)



Performance & Risk

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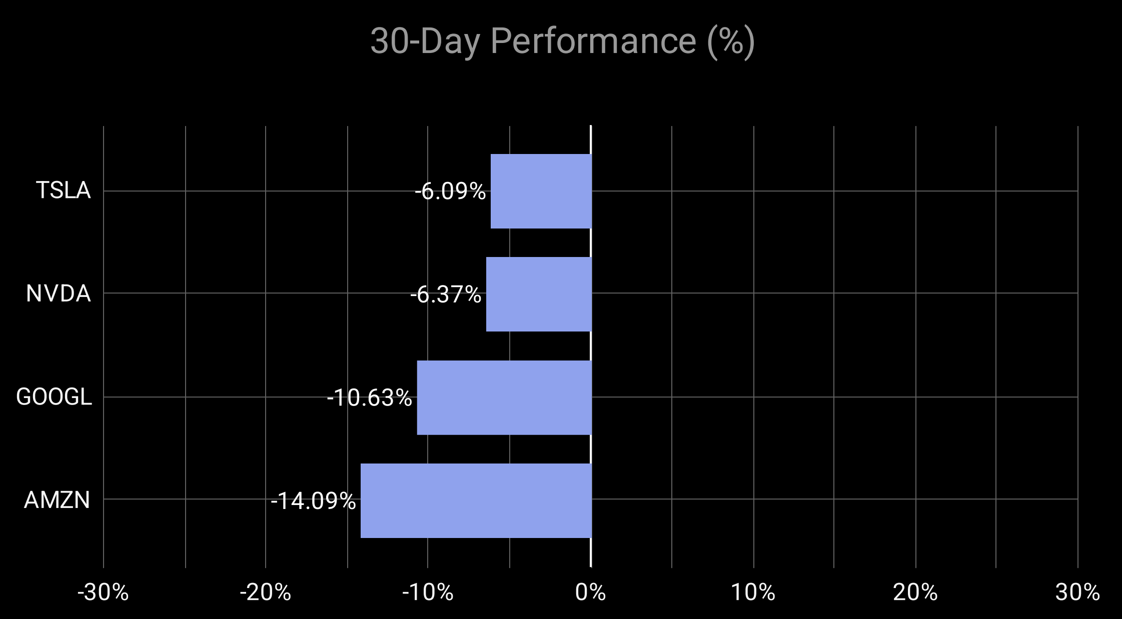

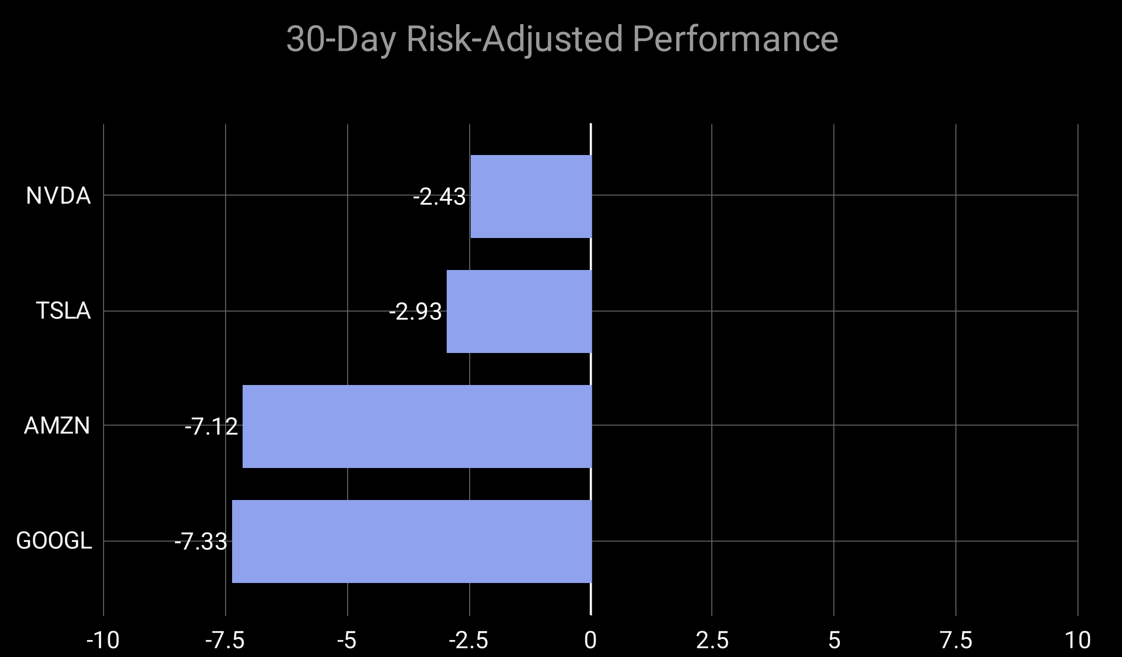

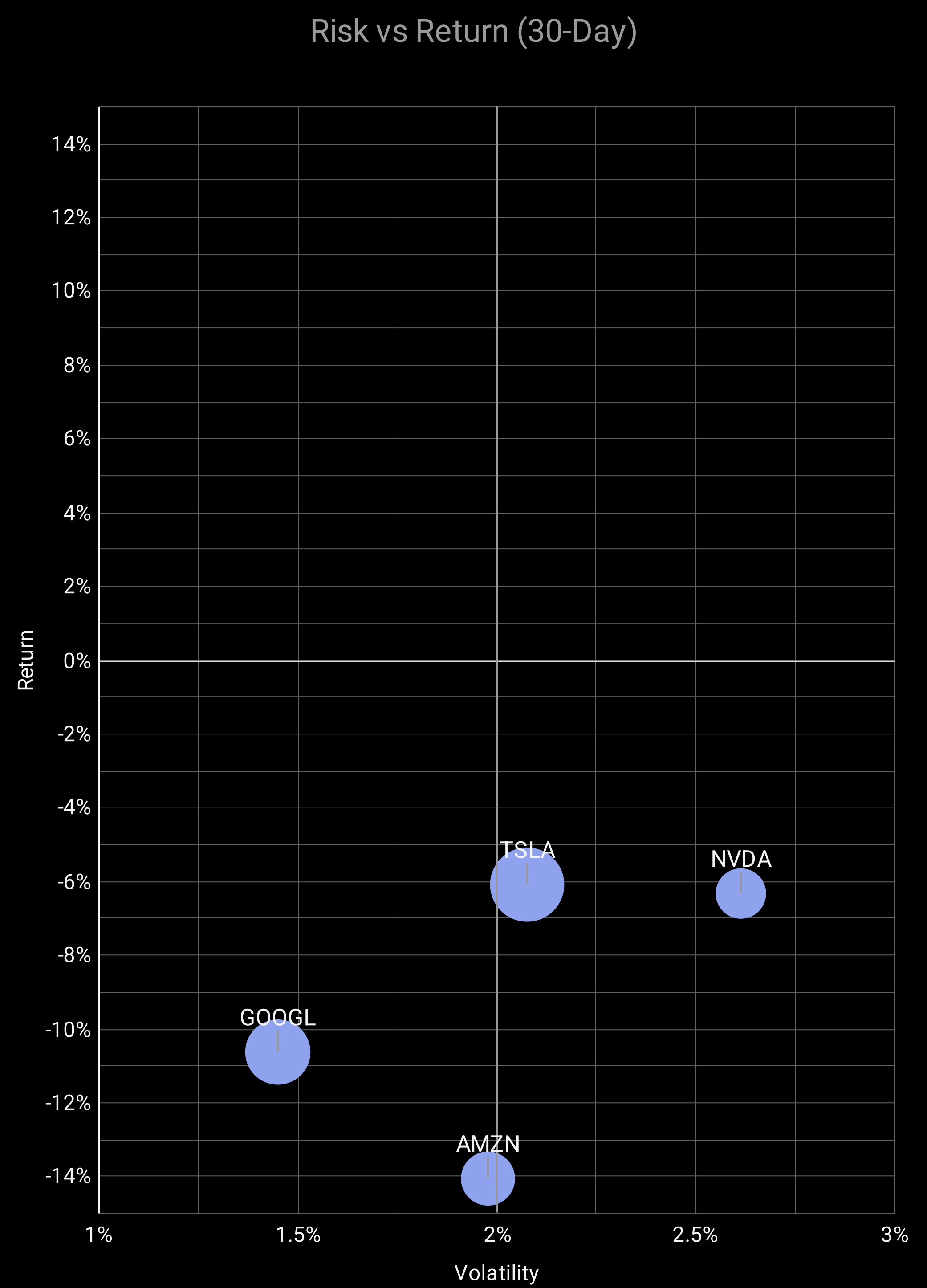

Portofolio & Diversification

30-Day Return Correlation

Ticker	TSLA	META	NVDA	GOOGL
TSLA	1.00	0.65	0.56	0.31
META	0.65	1.00	0.48	0.36
NVDA	0.56	0.48	1.00	0.25
GOOGL	0.31	0.36	0.25	1.00

Lower correlations indicate better diversification potential across the portfolio.